

$$\begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & 0 & -8 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row2}=\text{Row2}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+7}\times\text{Row2}]{\text{Row1}=\text{Row1}} \begin{bmatrix} 3 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\times 1/3]{\text{Row1}=\text{Row1}} \begin{bmatrix} 1 & 0 & 0 & 1/3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix} \therefore x = \begin{bmatrix} 1/3 \\ 4 \\ -6 \end{bmatrix} \square$$

$$3. A = \begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution: We shall have to solve the equation $Ly=b$ and $Ux=y$, where $L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. By Theorem 3 from Chapter 1, the matrix equation $Ly=b$ and $Ux=y$ has the same solution set as the system of equations corresponding to the augmented matrix $[L \ b]$ and $[U \ y]$.

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ -3 & 1 & 0 & 0 \\ 4 & -1 & 1 & 4 \end{bmatrix} \xrightarrow[\text{+3}\times\text{Row1}]{\text{Row2}=\text{Row2}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 4 & -1 & 1 & 4 \end{bmatrix} \xrightarrow[\text{-4}\times\text{Row1}]{\text{Row3}=\text{Row3}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow[\text{+Row2}]{\text{Row3}=\text{Row3}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \therefore y = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 2 & 1 \\ 0 & -3 & 4 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow[\text{-2}\times\text{Row2}]{\text{Row1}=\text{Row1}} \begin{bmatrix} 2 & -1 & 0 & -5 \\ 0 & -3 & 4 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow[\text{-4}\times\text{Row3}]{\text{Row2}=\text{Row2}} \begin{bmatrix} 2 & -1 & 0 & -5 \\ 0 & -3 & 0 & -9 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow[\times 1/3]{\text{Row2}=\text{Row2}} \begin{bmatrix} 2 & -1 & 0 & -5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow[\text{+Row2}]{\text{Row1}=\text{Row1}} \begin{bmatrix} 2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row1}=\text{Row1}} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\therefore x = \begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix} \square$$